Project Documentation

AI-Enhanced Customer Feedback Data Pipeline

Organization / Institution:	CLS Learning Solutions
Project Name:	AI-Enhanced Customer Feedback Data Pipeline
GitHub Repository:	https://github.com/your-org/ai-customer-feedback-pipeline
Team members:	Ahmed Sherif Youssef Mohammed Mohammed Mamdouh Youssef Essam Youssef Zeinhom Mahmoud Mohammed

1. Project Overview & Executive Summary

This project documentation establishes the blueprint for building an AI-Enhanced Customer Feedback Data Pipeline as part of the graduation requirements at CLS Learning Solutions. The pipeline addresses critical industry needs by transitioning raw, unstructured customer sentiment into structured actionable insights using modern data engineering workflows and pre-trained Natural Language Processing (NLP) models.

2. Problem Statement & Objectives

2.1 Problem Statement

Organizations daily receive thousands of unstructured customer reviews, support comments, and feedback items across various communication channels (CSV files, APIs, and databases). Business analysts face major bottlenecks trying to manually categorize and analyze these inputs, leading to delayed decision-making, missed sentiment trends, and unaddressed customer pain points.

2.2 Project Objectives

- **Data Ingestion:** Ingest unstructured customer feedback from multi-channel raw sources (CSV/API/Database).
- **Data Processing:** Implement robust data validation, deduplication, and text normalization procedures.
- **AI Enrichment:** Integrate a Hugging Face pre-trained AI model for automated sentiment analysis and tokenization.
- **Storage & Persistence:** Store both raw and enriched data records systematically in a structured relational database.
- **Orchestration:** Orchestrate end-to-end inferencing as an automated ETL transformation using Apache Airflow DAGs.
- **Reporting & Analytics:** Generate analytical aggregation summaries filtered by Product, Category, Date, and Sentiment.

3. Key Features & System Scope

- **Multi-source Ingestion:** Inbound pipeline configured for raw customer feedback ingestion from CSV files, web APIs, and relational databases.
- **Text Cleaning & Normalization:** Automated validation, removal of duplicate records, case normalization, and removal of noise/special characters.
- **Hugging Face AI Enrichment:** Tokenization of input sequences followed by sentiment classification, confidence/probability scoring, and processing timestamp generation.
- **Dual-layer Storage:** Persistence layer utilizing SQL/Pandas to maintain lineage for raw inputs alongside enriched outputs.
- **ETL Orchestration (Airflow):** Automated workflow orchestration using Apache Airflow, managing failure handling, retraining triggers, and task dependencies.
- **Optional LLM Extension:** Structured extraction using LLM/LangChain to extract entity tags such as product, issue_type, complaint, and requested_action.

4. Expected Deliverables (Part 1 & Final)

Deliverable Component	Description	Phase Status
Raw-Data Ingestion Module	Scripts and connectors to ingest data from CSV/API/Database.	Part 1 Ready
Text-Processing Module	Validation, tokenization, deduplication, and normalization functions.	Part 1 Ready
AI Transformation Module	Hugging Face pre-trained model inference pipeline.	Part 1 Ready

Database Schema & Storage	SQL relational schema design for raw & enriched records.	Part 1 Ready
Apache Airflow DAG	ETL DAG orchestration scripts for pipeline execution.	In Progress
Data Quality & Failure Checks	Validation assertions, null-handling, and failed inference handling.	In Progress
Analytical Visualization & Sentiment Dataset	Aggregated dataset and visual charts by Product/Category/Date.	In Progress
Architecture Diagram & README	System architecture diagrams and GitHub repository setup.	Part 1 Ready

5. Team Governance & Administration Guidelines

5.1 Team Composition & Roles

Per CLS Learning Solutions graduation project directives, the team must strictly adhere to the following formation rules:

- **Requirement 1:** Team Size: Total membership must strictly range between 4 and 6 trainees.
- **Requirement 2:** Team Leadership: A designated Team Leader must be explicitly assigned to lead coordination, documentation, and submissions.
- **Requirement 3:** Data Verification: All team members must verify their full details in the shared Google Drive Projects File.

5.2 Deliverables Timeline & Key Deadlines

- **Milestone 1 (Registration):** Thursday, October 08, 2026 — Finalize project name selection, initialize GitHub repository, and update the shared Google Drive Projects file.
- **Milestone 2 (Documentation Part 1):** Friday, October 16, 2026 — Submission of Part 1 Project Documentation compiled into a single consolidated PDF file.
- **Instructor Guidance:** Consult with the Technical Instructor for explicit technical clarifications and feedback loops prior to final defense.